

Second Year Diploma In Pharmacy

Hospital And Clinical Pharmacy

Dr. A. R. Paradkar S. A. Chunawala



 **NIRALI**
PRAKASHAN
ADVANCEMENT OF KNOWLEDGE

A TEXT BOOK OF

HOSPITAL AND CLINICAL PHARMACY

**FOR
SECOND YEAR DIPLOMA IN PHARMACY**

As per E. R. 1991

Dr. A. R. PARADKAR

M. Pharm., Ph.D.

Professor, Poona College of Pharmacy (Degree)

Pune 411 038.

Prof. S. A. CHUNAWALA

B. Com., M.B.A., D. Pharm

Ex. Reader in Business Management

Anand Commerce College,

Anand.

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PREFACE

Role of Pharmacist is growing rapidly with the community pharmacy services. Although in India this growth is very slow, introduction of **Hospital and Clinical Pharmacy** as a subject at **Diploma in Pharmacy** will significantly improve the approach of the upcoming pharmacists.

This book has been specifically written to meet the needs of Second Year Diploma Course in Pharmacy. The book gives elaborate and point to point explanation regarding different topics in Hospital and Clinical Pharmacy according to Education Regulations 1991. This is our sincere attempt to present the concepts in pharmacy in simple and lucid language.

With the developing interest and increasing potential in the field of Hospital and Clinical Pharmacy, we feel that this book will generate enough enthusiasm among the students, teachers and fellow Pharmacists and will offer a helping hand to the Practice of Pharmacy.

We wish to place on record our sincere thanks to **Mr. Dineshbhai Furia, Nirali Prakashan, Pune.**

We are always open to constructive criticism and timely suggestions from all strata of the profession in order to improve the quality in subsequent editions of this book.

AUTHORS

SYLLABUS

Part - I : Hospital Pharmacy

1. Hospitals - Definition, function, classification based on various criteria, organization, management and health delivery system in India.
2. **Hospital Pharmacy :**
 - (a) Definition.
 - (b) Functions and objectives of hospital pharmaceutical services.
 - (c) Location, layout, flow chart of materials and men.
 - (d) Personnel and facilities requirements including equipments based on individual and basic needs.
 - (e) Requirements and abilities required for hospital pharmacists.
3. **Drug Distribution System in Hospitals :**
 - (a) Out-patient services.
 - (b) In patient services : (i) types of services, (ii) detailed discussion of unit dose system, floor ward stock system, satellite pharmacy services, central sterile services, bed side pharmacy.
4. **Manufacturing :**
 - (a) Economical considerations, estimation of demand.
 - (b) Sterile manufacture - large and small volume parenterals, facilities, requirements, layout, production planning, man-power requirements.
 - (c) Non-sterile manufacture - Liquid orals, external, bulk concentrates.
 - (d) Procurement of stores and testing of raw materials.
5. Nomenclature and uses of surgical instruments hospital equipment and health accessories.
6. P.T.C. (Pharmacy Therapeutic Committee), hospital formulary system their organisation, functioning, functioning and composition.
7. Drug information service and drug information bulletin.
8. Surgical dressing like cotton, gauze bandages and adhesive tapes including their pharmacopoeial test for quality. Other hospital supply e.g. I.V. sets, B.C. sets, Ryal tubes, Catheters, Syringes etc.
9. Applications of computers in maintenance of records, inventory control, medication monitoring, drug information and data storage and retrieval in hospital and retail pharmacy establishments.

Part - II : Clinical Pharmacy

1. Introduction to clinical pharmacy practice - definition, scope.
2. Modern dispensing aspects - pharmacists and patient counselling and advice for the use of common drugs, medication history.
3. Common daily terminology used in the practice of medicine.
4. Disease, manifestations and pathophysiology including salient symptoms to understand the disease like Tuberculosis, Hepatitis, Rheumatoid Arthritis, Cardio-Vascular diseases, Epilepsy, Diabetes, Peptic Ulcer, Hypertension.
5. Physiological parameters with their significance.
6. Drugs interactions :
 - (a) Definition and introduction.
 - (b) Mechanism of drug interaction.
 - (c) Drug-drug interaction with reference to analgesics, diuretics, cardio-vascular drugs. Gastro-intestinal agent vitamins and hypoglycemic agents.
 - (d) Drug-food interaction.
7. Adverse drug reactions :
 - (a) Definition and significance.
 - (b) Drug-induced diseases and teratogenicity.
8. Drugs in clinical toxicity - Introduction, general treatment of poisoning systematic antidotes. Treatment insecticide poisoning heavy metal poison, Narcotic drugs, Barbiturate, Organophosphorus poisons.
9. Drug dependence, drug abuse, additive drugs and their treatment complications.
10. Bio-availability of drugs, including factors affecting it.

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∞ **Salient Features** ∞

- * Sufficient number of Tables are used to provide their comparative study in short.
- * Definitions of Pharmaceutical and Medical terms are given to understand the subject more suitably.
- * Good number of surgical instruments and equipments are described.
- * Three types of questions, at the end of the chapter i.e. short answered, long answered and objective, offer clue to the students to pay the attention to important contents.
- * Introduction of every chapter offers brief-account of the contents, to the students.

Part I - Hospital Pharmacy

Chapter ... 1

Hospitals and Their Organization

INTRODUCTION

Having now entered the 21th century, we are witnessing a spectacular consciousness of health and hospital services in India. In fact, many of us are born in hospitals and our association with them continues throughout our lives, resulting in an overall acceptance of hospitals as a part of our life.

There are some 6000 hospitals that dot the country. We have invested over ₹ 3,000 crores on them. Yet for a country which spends so generously for pleasure, we are almost tight-fisted when it comes to upgrade the overall effectiveness of our hospitals as organizations. Hospitals are greatly affected by technological advances in biological sciences including medicine, pharmacy and nursing. Herein lies a great challenge - they have to adapt continuously to the changing environment.

Early Hospitals

The modern hospital is a fairly recent phenomenon. The institutions which were fore-runners to the hospital, bore little resemblance to hospitals as they are now. The earlier hospitals were lodgings for the poor and the needy. Mainly the pilgrims and the wayfarers were housed in them. They were almshouses, and it is only recently that have cast off this image. Another motivating factor to establish hospitals was to accommodate leprosy patients.

Throughout the middle ages, the hospitals were run by monastic orders. The first hospitals in the New World were founded by the Spanish in Mexico City (1524) and French in Canada.

There was a general tendency to lump together the socially depressed, the physically handicapped, the socially unwanted, the financially weak and the sick.

During small-pox epidemics, hospitals mushroomed to take care of the victims and for treating them. These faded out when vaccinations came into existence.

The earlier hospitals thus incidentally became associated with the care of the sick but they were run for the homeless sick. They were considered a last resort, since the death rate was very high.

Evolution of the Modern Hospital

Only at the beginning of this century, the hospitals expanded their facilities to private patients over and above the homeless sick. In the latter part of the last century, laboratory examinations became an integral part of hospitals. Aseptic surgery and other advances in medical science made them safe. Introduction of diagnostic tools like X-ray, sonography etc. was another major advance. Anaesthesiology became a specialized and professional branch. Physicians accepted hospitals as institutions for diagnosis and treatment. Hospitals began catering to the rich as well as the poor. The philanthropic leanings of society and the welfare state concept of government, gave further fillip to hospital's modernisation.

In the last few years, hospitals started emphasising not only the therapeutic aspects of health care, but preventive aspects also. They, in a sense, started responding to the community's needs.

Definition : The hospital is defined as 'an institution of community health'. Its functions embrace the entire spectrum of medical care - prevention, diagnosis, therapy, rehabilitation, education and research. Its primary commitment is towards the patient. The epitaph of Pasteur rightly sums up that, it has to cure sometimes, alleviate often and comfort always.

The modern hospital, while discharging its functions, operates as an office, a hotel, a lodge and a medical care outfit. It also assumes the role of a major educational and social service institution. Most importantly, it has to remain a customer-oriented institution, since it is for them that it exists.

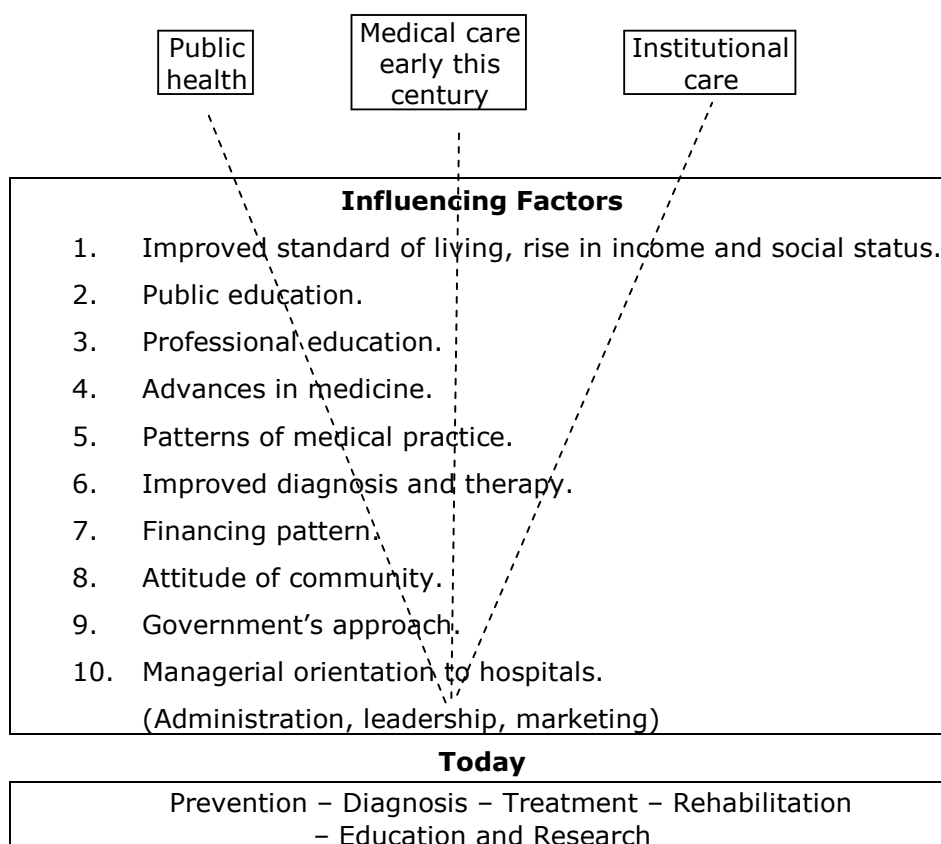
Functions of a Modern Hospital

A modern hospital is much more than a health care institution, since it exercises greater influence on the surroundings.

1. It raises the quality of care and general standards of medical practice.
2. It is a centre of community health and contributes a great deal to preventive and social medicine.
3. It lowers the incidence of disease through early detection and immediate treatment.

4. It is a link between the general public and policy-makers.
5. It stimulates the growth of medical science. Doctors and nurses receive their training in large teaching hospitals.
6. Bigger hospitals co-operate with smaller hospitals. Speciality hospitals co-operate with general hospitals. Private hospitals co-operate with public hospitals. The whole trend is towards integration.

The following diagram effectively illustrates the role of a modern hospital.



CLASSIFICATION OF HOSPITALS

Hospitals can be classified in a variety of ways. The most important criterion of classification is the clinical orientation of the hospital.

(1) Clinical Orientation as the Basis of Classification : Clinically, hospitals are classified on the basis of **major diseases** they seek to cure. Examples :

1. Psychiatric hospitals or Mental hospitals.
2. T.B. hospitals.
3. Leprosy hospitals.
4. Cancer hospitals.

They are also classified on the basis of their **anatomical-physiological specialisation**.

Examples :

1. Ear, nose and throat hospitals.
2. Orthopaedic hospitals.
3. Eye hospitals.
4. Kidney hospitals.

Further, on the basis of the **client group they serve**, they can be classified as :

1. Paediatric hospitals for children.
2. Gynaecological hospitals for women.
3. Maternity hospitals for mothers.

On the basis of **system of medicine** adopted for treatment, they can be classified as :

1. Allopathic (modern medicine) hospitals.
2. Ayurvedic (traditional system) hospitals.
3. Homeopathic hospitals.
4. Unani hospitals.
5. Nature cure centres.
6. Physio-therapy centres.

(2) Non-Clinical Basis of Classification :

(i) On the basis of ownership : Hospitals can be classified on the basis of ownership either as government or public (sector) hospitals and private (sector) hospitals. State-run hospitals (public hospitals) constitute 80% of hospital population.

Ownership

Public Ownership : They can be : Central Government Hospitals Like the Railway Hospitals, Defence Hospitals, All India Institute of Medical Sciences, P.G. Institute of Medical Sciences, Pondicherry or Chandigarh State Government Hospitals Like Civil Hospitals at district headquarters; Sassoon Hospital, Pune; J.J. Hospital, Mumbai, Local Self-Government Hospitals These are run by Municipalities or Corporations, e.g. BMC Hospitals like Bhagwati Hospital in Mumbai.	Private Ownership : They can be run by : Trusts The Board of Trustees manage the affairs, e.g. Mumbai Hospital, Mumbai; Jaslok Hospital, Mumbai. Religious Bodies/Orders Ram Krishna Hospital, Kolkatta; Christian Medical College Hospital, Vellore. Limited Companies They can be incorporated as public limited company, where the public subscribes to the share capital. e.g. Apollo Hospital, Ltd., Chennai; Medinova Centre's proposed limited hospital at Baroda, proposed limited hospital at Kanpur.
Hospitals → (Size based on number of beds) → (a) Large Hospitals (1000 and above) → (b) Medium Hospitals (500 to 1000) → (c) Small Hospitals (100 to 500) → (d) Very small Hospitals (Less than 100)	Other Private Hospitals/Nursing Homes These are operated in practically all the metros and towns in India by either a single or a group of private practitioners or husband-wife team. They are proprietary or partnership concerns and are general nursing homes, with facilities for general surgery. Some nursing homes are specialised e.g. paediatric or orthopaedic or ophthalmic. There is a demand to regulate the private nursing homes by a suitable enactment.

(ii) On the Basis of Size : Hospitals can be classified on the basis of size :

- (a) Large Hospitals :** (Beds 1000 and above), e.g. J.J. Group of Hospitals in Mumbai have an intake of 1400 beds; K.E.M., Mumbai has 1600 beds.
- (b) Medium Hospitals :** (Beds between 500 - 1000), e.g. Bombay Hospital, Mumbai has 700 beds, Jaslok Hospital, Mumbai has 320 beds.
- (c) Small Hospitals :** (Beds between 100 - 500) Breach Candy Hospital, Mumbai has a bed capacity of 130 beds (They intend to raise it by another 55 beds). P.D. Hinduja National Hospital, Mahim, Mumbai has a bed capacity of 175 beds.

(d) Very small Hospitals : (Beds less than 100).

(iii) On the Basis of Cost :

(a) Elite Hospitals like Jaslok and Hinduja are meant for the privileged. They are a symbol of high-tec medical development. The room rates vary between ₹ 300 - ₹ 1200 per day. The deluxe rooms are equipped with fridge, T.V. and telephone. Excepting the medical care, they are like five-star hotels. They are, therefore, called five-star hospitals. But even these elite institutions reserve a particular percentage of their capacity for the poorer sections and subsidise a particular percentage of their accommodation cost. Jaslok has reserved 25% for the poorer sections and 30% at half the rates. Out of 680 beds of Mumbai Hospital, 315 are free and 112 are subsidised.

(b) Budget Hospitals : These are meant for moderate budget and low budget users, e.g. civil hospitals, corporation hospitals etc.

Miscellaneous

Hospitals can be teaching hospitals, to which a medical college is attached. They can be research centres. They can be integrated hospitals, where many medical and surgical departments are integrated or alternative systems of medicine are integrated. There are in-plant hospitals, run as a part of business organisation for the benefit of its employees and immediate community surrounding it. There is no system of accreditation for hospitals here in India, as it prevails in other advanced countries. The time has come now to take steps for the rating of hospitals on the criteria of service quality, service availability, patient satisfaction and perceived image.

ORGANISATION OF A HOSPITAL

A successful hospital is based on a triad - good community-oriented planning, good design and construction and good administration. A progressive hospital builds its services on specific knowledge of the community it is to serve. Organisation is not viewed merely as a structure these days. It is a process of achieving the objectives by grouping people in order to get the work done.

The ultimate aim of the hospital is to provide optimum health care, and as such its organization is based upon the following principles :

(a) Team Approach : A patient here is under the care of a professional team of medical/paramedical staff. The services are integrated and co-ordinated.

(b) Spectrum of Services : As Drucker, the well-known management Guru observes, "activities analysis is an important criterion to shape the organization". Hospital, as an organization, has to undertake a number of activities - diagnosis, treatment or therapy, rehabilitation, education and prevention. These activities are further sub-classified, and groups and sub-groups are formed to carry out these activities. The whole workload is divided into manageable units of health services.

Each unit contributes to the total spectrum of health services that are feasible in terms of the type of community it serves, and the overall pattern of health facilities of the region in which it exists.

(c) Continuum : A hospital caters to both ambulatory and non-ambulatory patients (outpatients and inpatients). The organization should form a continuum with common or integrated services.

(d) Evaluation and Research : Services provided by the hospital should be rated in terms of quality and adequacy for meeting the patient's and community's needs.

(e) Authority-responsibility : The various people must be allotted carefully spelled out tasks in a hospital. Responsibility should be assigned appropriately. Once responsibility is assigned, accountability to report the superior regarding performance, is a natural corollary. This is necessary to ensure a high quality of patient care.

(f) Talent Search : A good hospital organization should be staffed by competent medical and non-medical personnel. The organization should formulate a programme to attract good physicians/surgical specialists to maintain optimum occupancy of inpatient facilities and full utilization of outpatient facilities.

(g) Budgeting and Financing : Large hospitals like K.E.M., Mumbai treat almost 15 lacs patient per year, and perform 18 thousand major surgeries and 33 thousand minor ones. The yearly budget for K.E.M. is ₹ 16 - 18 crores. Many other hospitals face a resource crunch. They should therefore design an effective organization, and use proven management techniques to optimize their resources. There should be scientific budgeting and a planned programme for capital financing.

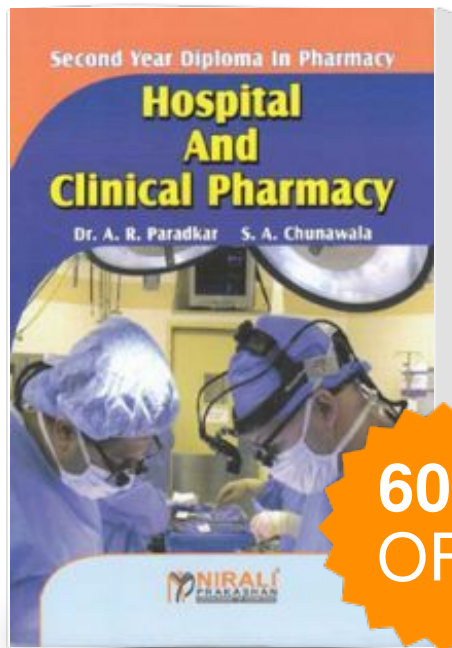
(h) No Stereotypic Organization Chart : There is no standardized organizational chart which will be suitable for every hospital. The organization should evolve considering the objectives, the tasks and their feasibility of the specific hospital.

(i) Governing Body : Each hospital has its top management, which directs its course of action, and which provides policy guidelines and exercises overall supervision and control. It assumes the legal and moral responsibility for the conduct of the hospital's staff as an institution. It is responsible to the patient, the community and the sponsoring organization.

A TYPICAL HOSPITAL ORGANISATION

A hospital, by its very nature, is a physical plant created for the purpose of health care of the patients. Since, its mission is health care, the functional department of clinical services is the most important division of a hospital. However, the clinical services need the support of services like pharmacy and nursing, and administrative support. Departments of pharmacy, nursing and office, and accounts administration are commonly found in all hospitals. The hospital, being a physical plant, needs

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Author : **Dr. A. R. Paradkar,**
Prof. S. A. Chunawala

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